

Assignment 1: Frontend Integration & Client-Side CRUD – Explanation

1. Project Overview

This project is a News Editor, a functional document management web application built with HTML, CSS, and JavaScript. It implements complete client-side CRUD operations using localStorage for data persistence, along with simulated AI writing assistance tools. The application features a split-panel layout with a document list sidebar and a full-featured text editor.

2. File Structure & Architecture

- index.html - Semantic HTML with ARIA attributes and responsive layout
- css/styles.css - CSS custom properties, dark mode, mobile-first responsive design (600+ lines)
- js/main.js - Three modular service components: DocumentService, AIService, UIController

The architecture uses a service layer pattern that abstracts data operations behind clean interfaces. DocumentService handles all CRUD via localStorage, AIService provides simulated AI functions, and UIController manages DOM interactions. This design enables easy replacement of localStorage with a real backend API in future assignments.

3. CRUD Implementation

Create: Click "New Document" to create a document with auto-generated unique ID and timestamps. Stored in localStorage as a JSON array and immediately opened in the editor for editing.

Read: Documents displayed in the sidebar as cards with title, category badge, excerpt, word count, and date. Click to open in the editor. Search bar and category filters provide quick discovery.

Update: Edit title, category, and content in the editor. Changes tracked with "Unsaved changes" status. Save via button or Ctrl+S. The updatedAt timestamp refreshes automatically on each save.

Delete: Protected by a confirmation modal showing the document title. On confirm, the document is removed from localStorage, editor closes, and a toast notification confirms deletion.

4. AI Writing Tools (Simulated)

No real API required. Both AI features are pure JavaScript with simulated delays (1-2.5s) mimicking real API behavior.

Improve Writing: Applies 30+ text transformations - vocabulary upgrades ("good" to "excellent"), redundancy removal, and structure enhancements. Shows improved text with change count in a result panel with "Apply to Editor" / "Dismiss" buttons.

Summarize: Extracts the topic sentence and key sentences containing importance indicators ("significantly", "demonstrate", "notably"). Produces a summary with word count stats and reduction percentage for review before applying.

5. Additional Features

Dark mode with persistence, real-time search and category filtering, responsive mobile design with collapsible sidebar, keyboard shortcuts (Ctrl+S), toast notifications, and 6 pre-loaded sample documents.

6. Future Assignment Readiness

DocumentService abstracts all data operations, enabling straightforward replacement of localStorage with HTTP API calls for Assignment 2 (Backend CRUD). AIService can be enhanced for Assignment 3 (AI Integration) to call real AI APIs while maintaining the same UI interaction pattern.